

SE (CS)
Spring Semester 2026

**Database Management Systems
(CS-222)**

*Lectures #3-5
An Architecture for a database system
Functions of DBMS*

*Dr Syed Zaffar Qasim
CIS Dept*

1

Three-Level Architecture of a Database System

- The 3-level architecture was proposed by Standards Planning and Requirements Committee (SPARC) of ANSI in mid 1970s.
- The three levels in this *ANSI/SPARC model* are
 - the *conceptual level*,
 - the *external level* and
 - the *internal level*.
- It provides a basis for understanding some of the functionality of a DBMS.
- The architecture also referred to as *Three-Schema Architecture*.

2

Objectives Of Three-Level Architecture

1. Abstract View of Data

- For efficient data retrieval, *complex data structures* and efficient *access techniques* (indexing or hashing) are used.
- Since many database systems users are not computer trained
 - so a major purpose of a DBMS is to provide them with an *abstract view* of the data
 - by hiding certain details of how data is stored and manipulated.
- The complexity is hidden from them through *several levels of abstraction*
 - in order to simplify their interaction with the system.

3

Objectives Of Three-Level Architecture

2. Achieve Data Independence

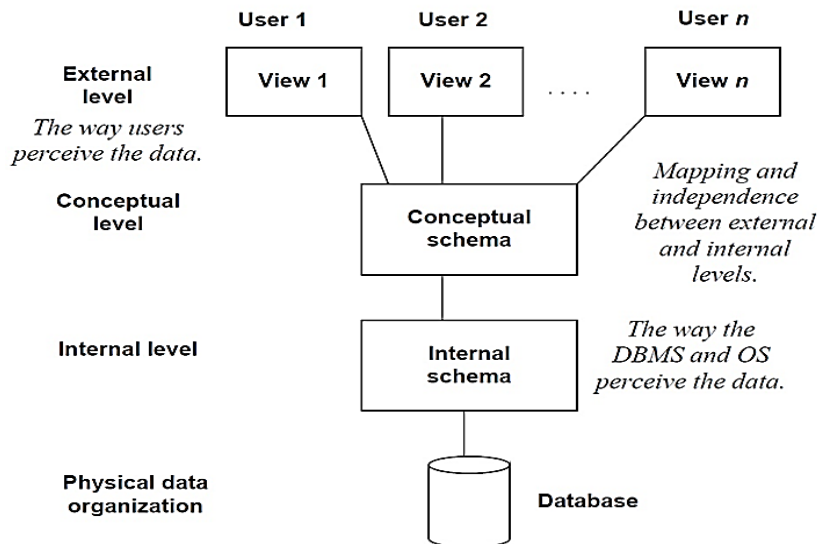
- *Data Independence: The immunity of applications to changes in physical representation and access techniques of data.*
- Made possible by separating the data descriptions (metadata) from application programs that use the data.
- The three-level architecture provides a basis for achieving the data independence objective.
- The DBA should be able to change the database storage structures without affecting the users' view.

3. Customized Access

- Each user should be able to access the same data but have a different customized view of that data.

4

The Three Levels of Architecture



5

Conceptual Level

- Also known as the *community logic level* or sometimes just the *logic level*, unqualified.
- At the conceptual level, we have an *integrated view* of the database called *conceptual view*.
- It is the *level of indirection* between the other two levels.
- In the relational model, the *base tables* correspond to the ANSI/SPARC conceptual level.
- The conceptual view is the *view of the database administrator* who must decide what information is to be kept in the database.

6

Conceptual Level Conceptual View

- The conceptual view represents the *entire* database in an *abstract* form in comparison with the way it is physically stored.
- The conceptual view consists of *many occurrences* of each of many types of conceptual record.
- For example, it consists of
 - a collection of Customer record occurrences plus
 - a collection of Order record occurrences plus
 - a collection of Payment record occurrences (etc., etc.).
- A *conceptual record* is not necessarily the same as either an *external record* on the one hand, or an *internal (or stored) record* on the other.

7

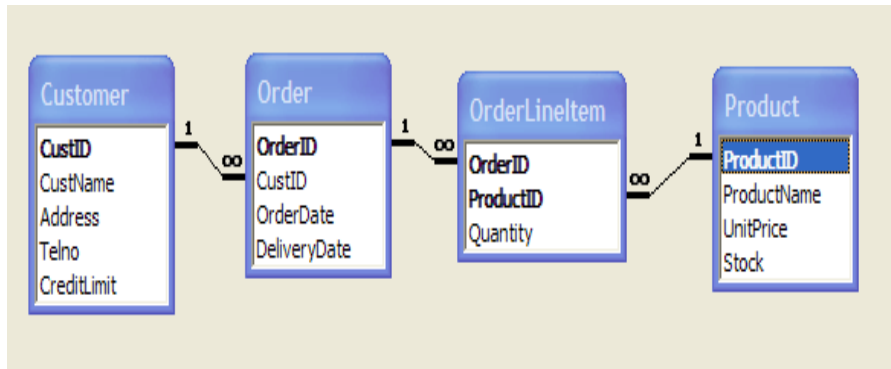
Conceptual Level Conceptual Schema

- The conceptual view is defined by means of the conceptual schema.
- A conceptual schema is a detailed specification of the entire organizational data, while being *independent* of any database management technology.
- The conceptual schema includes following requirements:-
 - All the entities, their attributes and their relationships.
 - Definitions of each of the various conceptual record types that represent the entities, the data item types that represent their attributes and the relationships that will be stored.
 - Integrity constraints,
 - Security constraints,
 - Semantic information about the data meanings.

8

Conceptual Schema Using ER Model

- Conceptual schema can be depicted in graphical format using *ER or Object Modeling notations*;
- This type of conceptual schema is usually called a data model.



9

Conceptual Schema Using DDL

```

CREATE TABLE CUSTOMER
(CUSTID NUMBER(4) PRIMARY KEY,
CUSTNAME VARCHAR2(25) NOT NULL,
ADDRESS VARCHAR2(50),
TELNO VARCHAR2(20),
CREDITLIMIT NUMBER(6) CHECK
(CREDITLIMIT BETWEEN 0 AND 100000));
    
```

```

CREATE TABLE ORDER
(ORDERID NUMBER(4) PRIMARY KEY,
CUSTID NUMBER(4),
ORDERDATE DATE,
DELIVERYDATE DATE,
FOREIGN KEY (CUSTID)
REFERENCES CUSTOMER (CUSTID));
    
```

```

CREATE TABLE ORDERLINEITEM
(ORDERID NUMBER(4),
PRODUCTID NUMBER(4),
QUANTITY NUMBER(5),
PRIMARY KEY (ORDERID, PRODUCTID),
FOREIGN KEY (ORDERID)
REFERENCES ORDER(ORDERID),
FOREIGN KEY (PRODUCTID)
REFERENCES PRODUCT(PRODUCTID));
    
```

```

CREATE TABLE PRODUCT
(PRODUCTID NUMBER(4) PRIMARY
KEY,
PRODUCTNAME VARCHAR2(25) NOT
NULL,
UNITPRICE NUMBER(4),
STOCK NUMBER(4) CHECK
(STOCK BETWEEN 0 AND 1000));
    
```

10

External Level

- External level, also the *user logical level*,
 - is the one concerned with the way the data is seen by individual users.
- Comprises several *views* of individual users or application programmers.
- Derived from the conceptual view.
- Each external view describes only part of the entire database that the user is interested in.
- In the relational model, *views* correspond to the ANSI/SPARC external level.

11

EXTERNAL LEVEL

- Different views may have different representations of the same data.
- e.g., one user may believe dates are stored in the form (*month, day, year*),
 - while another may believe they are represented as (*year, month, day*).
- Some views might include *virtual or calculated data*.
- For example, *age* may not be actually stored, but *date of birth* may be, and
 - *age* may be calculated by the system when the user refers to it.
- Also, data calculated from *several* records.

12

External Level

- Each external view is defined by means of an *external schema* (also called *subschema*).
- External Schema consists basically of
 - definitions of each of the various external record types in that external view.
- The schemas are compiled by the DBMS and
 - stored in object form for use by the data dictionary in retrieving records.
- They should also be kept in source form as documentation.

13

External Schema using DDL

<pre>CREATE VIEW ORDER_VW AS SELECT ORDERID, CUSTID CUSTNUM, (DELIVERYDATE - ORDERDATE) DAYS_FOR_PROCESSING FROM ORDER;</pre>	<pre>CREATE VIEW CUSTOMER_ORDER_VW AS SELECT ORDERID, CUSTNAME, ORDERDATE, DELIVERYDATE FROM CUSTOMER, ORDER WHERE CUSTOMER.CUSTID = ORDER.CUSTID; // using inner join</pre>
---	--

```
SELECT * FROM ORDER_VW;  SELECT * FROM CUSTOMER_ORDER_VW;
```

14

Internal Level

- The internal level covers the *physical implementation of the database* to achieve optimal run-time performance and storage space utilization.
- It considers the *data structures* and *file organizations* used to store data on storage devices.
- This level is the *responsibility of physical database designers/database administrator* who decide
 - which physical devices will contain the data,
 - what access methods will be used to retrieve and update data and
 - what measures will be taken to maintain or improve database performance.

15

Internal Schema

- The internal schema comprehensively defines the whole database using a *technology-dependent style*.
- The DBMS chosen determines what structures are available.
- The internal schema is concerned with such things as
 - *Complex low-level data structures and file organizations used to store the data on physical storage devices*
 - *Indexing and Hashing schemes used*
 - *The methods of representation of data fields*
 - *Physical sequence of records*
 - *Data compression and data encryption techniques*
- The internal schema is written using DDL extensions on underlying DBMS.

16

MAPPING

- *Correspondence between two levels* in three-schema architecture of a database system.
- **Benefit:** *provides isolation* between the different levels.
- There are two levels of mapping in the architecture,
 - one from the external level to conceptual level and
 - one from the conceptual level to the internal level.
- The *external/conceptual mapping* tells the DBMS
 - which objects on the conceptual level correspond to
 - which objects in a particular user's external view.
- There may be differences in record names, data item names, data item order, data types and so on.

17

External/Conceptual Mapping

- Moreover, several conceptual fields can be combined into a single (virtual) external field, and so on.
- If changes are made to either an external view or the conceptual model, the mappings must be changed.
- The external/conceptual mapping is the key to *logical data independence*.
- **Logical data independence:** the immunity of external schemas to change in the conceptual schema.

18

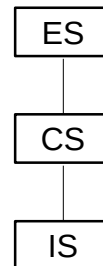
Conceptual/Internal Mapping

- The *conceptual/internal mapping* defines the correspondence between the conceptual objects and internal ones.
- It specifies how the conceptual records and fields are represented at the internal level.
- If the structure of the stored database is changed –
 - then conceptual/internal mapping must be changed accordingly, so that the conceptual schema can remain invariant.
- The conceptual/internal mapping is the key to *physical data independence*.
- *Physical data independence*: the immunity of the conceptual schema to the changes in the internal schema.

19

Database Development using Three-Schema Architecture

- External schemas not necessarily developed before the conceptual schema.
- The conceptual and external schemas are usually developed iteratively.
- Often we develop a first cut at the conceptual schema based on the general understanding of database requirements on a project.
- Then we develop external schemas (user views) for each transaction, report, screen display etc.
- Finally we write the specifications for the associated internal schema.



20

Mechanism for Data Retrieval in DBMS

1. A user issues an access request, using some particular data query language (typically SQL)
e.g. **SELECT * FROM ORDER_VW;**
2. The DBMS intercepts that request and analyzes it
3. The DBMS inspects, in turn,
 - the external schema for that user,
 - the corresponding external/conceptual mapping,
 - the conceptual schema,
 - the conceptual/internal mapping and
 - the storage structure definition.
4. The DBMS executes the necessary operations on stored database.

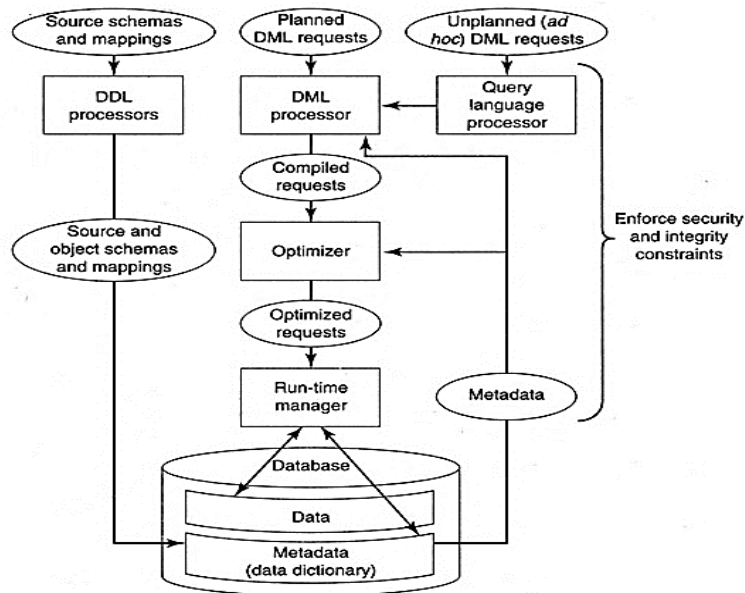
21

Retrieval of an external record occurrence

- For accessing an external record occurrence,
 - *fields will be required* from several conceptual record occurrences and
 - each conceptual record occurrence in turn will *require fields* from several stored record occurrence.
- Conceptually, then, the DBMS must first *retrieve* all required stored record occurrences,
 - then *construct* the required conceptual record occurrence
 - and then *construct* the required external record occurrence.
- At each stage, data type or other conversions might be necessary.
- This implies poor performance because of run-time overhead.
- In practice, however, it might be possible for access requests to be compiled prior to run time.

22

Major DBMS functions and components



23

Functions of DBMS

1. **Data definition**
 - Using DDL
 - For tables, views, indices etc.
2. **Data Manipulation**
 - to handle requests to retrieve, update or delete existing data or to add new data to the database.
3. **Integrity enforcement**
4. **Security enforcement**
5. **Interaction with file manager**
6. **Optimization and Execution**
7. **Backup and Recovery**
8. **Concurrency Control**
9. **Data Dictionary**

24

Data definition

- The DBMS must be able to accept data definitions
 - i.e. external schemas, the conceptual schema, the internal schema and all associated mappings in *source form* and
 - convert them to the appropriate *object form*.
- The DBMS must include DDL processors for this conversion.

Data Manipulation

- The DBMS must be able to handle requests to *retrieve, update* or *delete* existing data or to *add* new data to the database.
- The DBMS must include a *DML processor* or *DML compiler* component to deal with the data manipulation language (DML).

25

Types of DML requests

Planned Request

1. One for which the need was *foreseen* well in advance of time the request is executed.
2. The DBA will probably have *tuned* the physical database design to guarantee good performance for such requests.
3. Characteristic of *operational or production applications*.
4. Will typically be issued from prewritten application programs.

Unplanned Request

1. An adhoc query, i.e., the need was *not seen in advance*, but instead arose in a spur-of-the-moment fashion.
2. The physical database design might or might not be ideally suited for the specific request under consideration.
3. Characteristic of *decision support applications*.
4. Will be issued interactively via some query language processor.

26

Integrity Enforcement

- The database manager can then determine whether updates to the database results in violation of the constraint;
 - if so, then appropriate action must be taken.
- These checks are normally administered through the data dictionary.

Security Enforcement

- Not every database user needs to have access to the entire content of the database.
- It is the job of the database manager to enforce these security requirements.

27

Interaction with the file manager

- The database manager translates the various DML statements into low-level file system commands.
- The file system commands are then passed on to the file manager for updating the data files.

Backup and recovery

- A computer system failures include disk crash, power failure and software errors causing loss of database.
- It is the responsibility of the database manager to *detect* such failures and *restore* the database to a state that existed prior to the occurrence of the failure.
- This is usually accomplished through the initiation of various backup and recovery procedures.

28

Concurrency Control

- When several users update the database concurrently, the consistency of data may no longer be preserved.
- Controlling the interaction among the concurrent users is another responsibility of the database manager.

Data Dictionary (DD)

- The DD is a repository of all definitions of data (sometimes called metadata) used by the organization.
- DD can be regarded as a database in its own right (but a system database, rather than a user database).

29

Information normally catalogued in the DD

1. *Description of each record:* record name, size, component data items.
2. *For each data item type:* data item name, length, type and a brief narrative description.
3. Identification of primary and foreign keys.
4. All the various schemas and mappings (external, conceptual, etc.) will physically be stored, in both source form and object form, in the dictionary.
5. Integrity constraints
6. Security information

30

Information normally catalogued in the DD

- It is possible to query the dictionary just like any other database.
- In Oracle DBMS, some of the available views are
 - USER_TABLES
 - USER_CATALOG
 - USER_INDEXES
 - USER_CONSTRAINTS
 - DBA_USERS

31

Database Administrator (DBA)

- *Person responsible for physical database design and implementation, setting security and integrity controls, database performance monitoring and backup/recovery associated with managing a database.*

1. Defining the Conceptual Schema

2. Defining the Internal Schema

- CREATE TABLE XYZ (..., **PCTFREE 15**);
 - ❖ Amount of space reserved in each disk block of table for future updates of existing rows.

3. Liaising with users

- It is the business of DBA to liaise with users to ensure that
 - ❖ the data they need is available and
 - ❖ to write (or help the users write) the necessary external schemas, using the applicable DDL.

4. Defining Dump and Reload policies

32